



Qualcomm Linux AI/ML Guide

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Contents

1	Overview	3
2	Architecture	5
2.1	AI hardware accelerators	6
2.2	AI software stack	7
3	APIs	8
4	Sample Apps	10
4.1	AI processing for object classification	10
4.2	AI processing for object detection	11
4.3	AI processing for multi-input/output object detection	11
4.4	Parallel AI fusion	11
4.5	AI processing for pose detection	12
4.6	AI processing for image segmentation	12
5	Developer workflow	13
6	References	14

1 Overview

Qualcomm® Linux AI stack allows developers to optimally deploy pre-trained, deep learning models on Qualcomm hardware accelerators, such as Neural Processing Unit (NPU), Graphic Processing Unit (GPU), and Central Processing Unit (CPU). Qualcomm AI software offering contains software development kits (SDKs), APIs, sample applications, development tools, and third-party frameworks support such as GStreamer and TFLite, to ease application development.

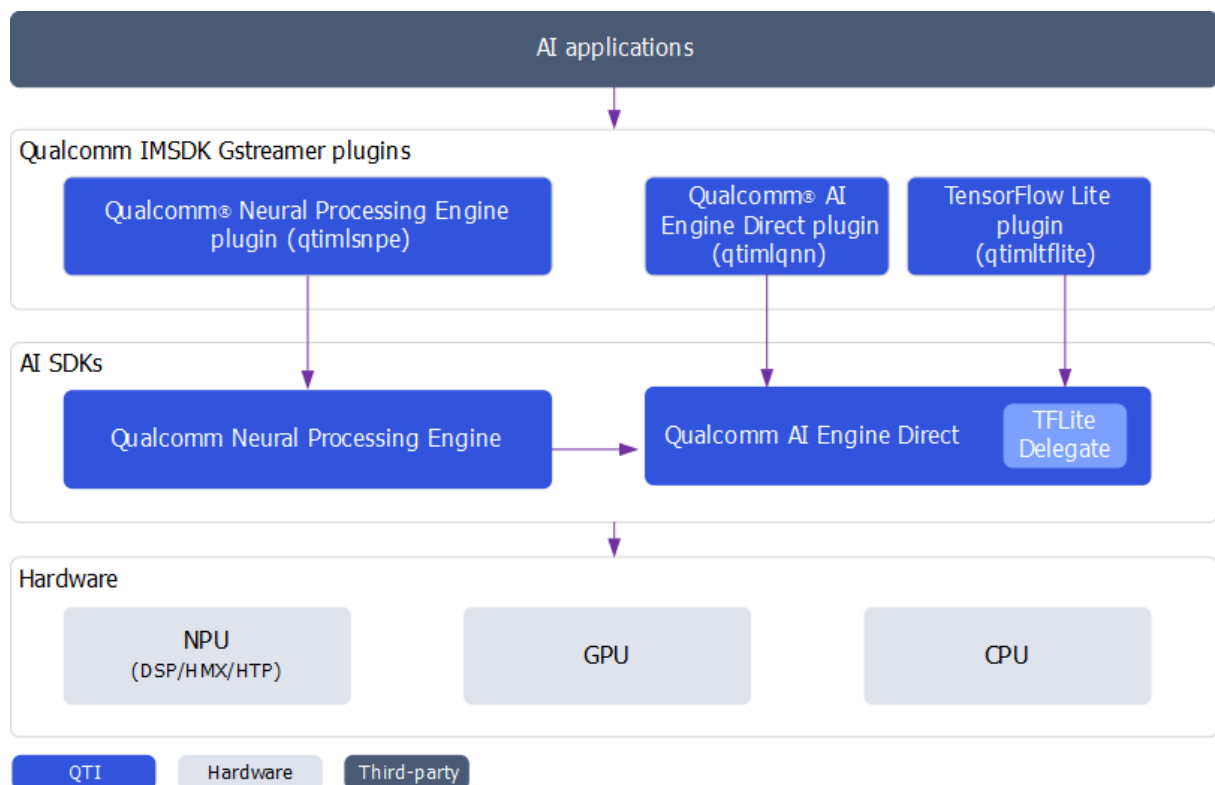


Figure1 Top-level AI hardware and software blocks of the Qualcomm Linux AI stack

The key components of the Qualcomm Linux AI stack are:

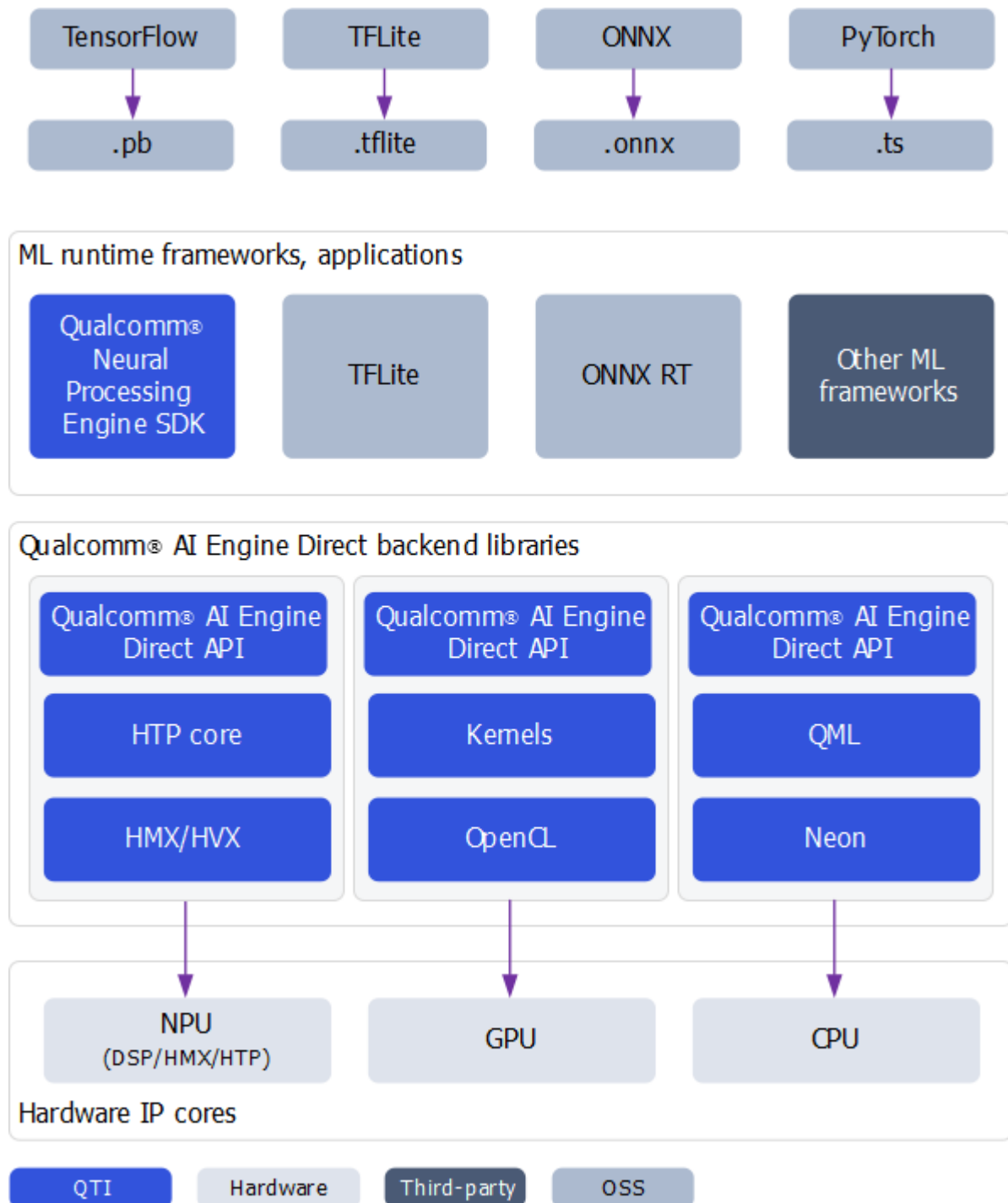
- **AI applications** - Sample applications based on GStreamer that can be used or customized as needed.
- **Gstreamer plugins** - Qualcomm Linux software offers GStreamer-based, machine learning

plugins for accelerating AI inference using TFLite, Qualcomm® Neural Processing Engine SDK, etc., along with Gstreamer plugins for pre- and post-processing.

- **Qualcomm AI Stack** consists of two SDKs to accelerate AI workloads. The **Qualcomm Neural Processing Engine SDK** and **Qualcomm AI Engine Direct** provide tools, libraries, etc., to optimally accelerate AI models on multiple hardware accelerators.
- Qualcomm SoCs offer three **hardware cores** for AI loads.
 - **Neural Processing Unit (NPU)** - Also referred to as Qualcomm® Hexagon™ Tensor Processor (HTP) or DSP/HMX, is suitable for executing AI workloads with low-power and high-performance. To get optimized performance, pre-trained models need be quantized to one of the supported precisions.
 - **Graphics Processing Unit (GPU)** - Qualcomm® Adreno™ GPU is suitable for executing AI workloads with medium-power, and medium-performance. AI workloads are accelerated with OpenCL kernels. The GPU can also be used to accelerate model pre/post processing.
 - **Central Processing Unit (CPU)** - AI inferencing on the CPU can be used to benchmark model accuracy/performance against other hardware accelerators. The CPU can also be used to run model pre/post processing.

2 Architecture

The Qualcomm AI offering consists of hardware accelerators and AI SDKs to harness the power of hardware.



2.1 AI hardware accelerators

AI workloads can be accelerated on multiple hardware cores:

- Qualcomm® Hexagon™ Tensor Processor (HTP) - Also known as NPU/DSP/HMX, suitable to execute AI workloads with low-power and high-performance. For optimized performance, pre-trained models need be quantized to one of the supported precisions.
- Qualcomm® Adreno™ GPU - Suitable to execute AI workloads with medium-power, and medium-performance. AI workloads are accelerated with OpenCL kernels. The GPU can also be used to accelerate model pre/post-processing.
- Qualcomm® Kryo™ CPU - AI inferencing on CPU can be used to benchmark model accuracy/performance against other hardware accelerators. The CPU can also be used to run model pre/post processing.

2.2 AI software stack

AI stack contains SDKs to harness the power of AI hardware accelerators. Developers can use the stack of their choice to deploy AI workloads. Pre-trained models (with the exception of TFLite models) need to be converted to an executable format with the selected AI Stack SDK before running them. Note that TFLite Delegate allows developers to directly run TFLite models.

- [Qualcomm Neural Processing Engine \(SNPE\)](#)

An all-in-one SDK that provides C, C++, and Java APIs to support heterogeneous computing, system-level configurations, and direct AI workloads to all accelerator cores. Provides developers with flexibility, including inter-core collaboration support and other advanced features.

- [Qualcomm AI Engine Direct \(QNN\)](#)

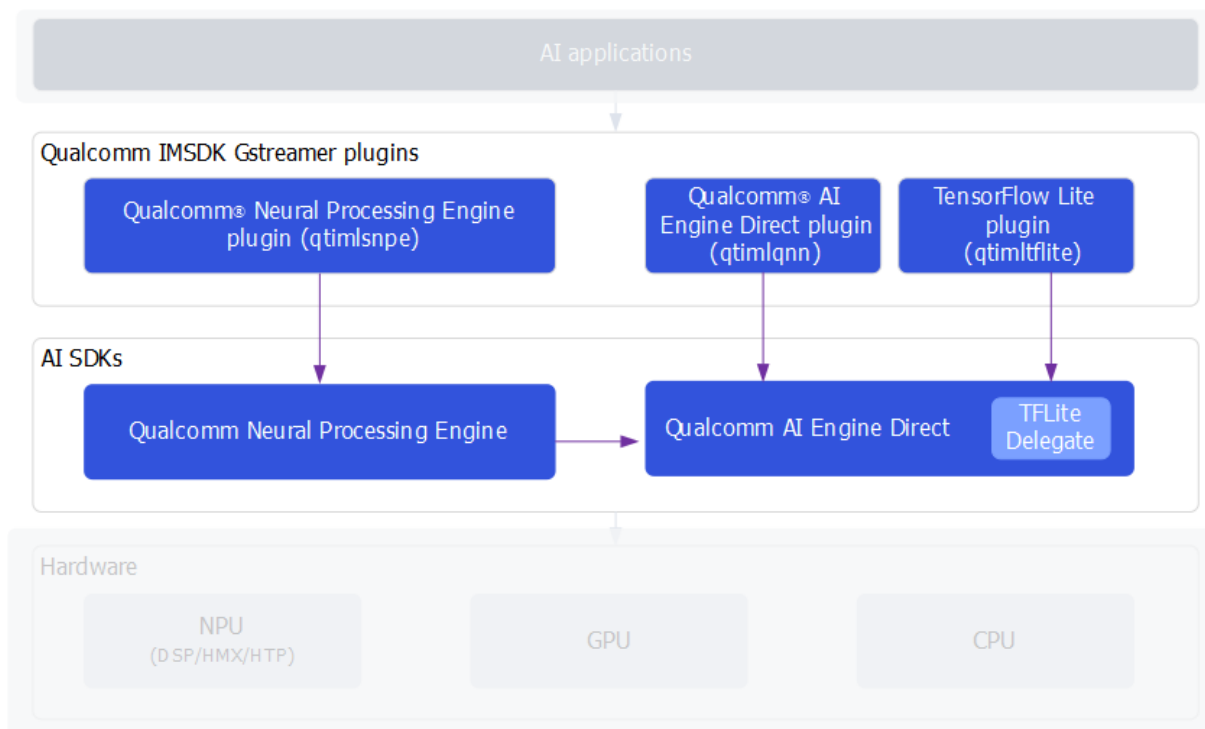
Lower-level, highly customizable unified APIs that speed up AI models on all AI accelerator cores with individual libraries. Can be used directly to target a specific accelerator core or delegate workloads from popular runtimes including Qualcomm Neural Processing Engine SDK, TensorFlow Lite, and ONNX runtime. Low-level SDK provides more functionality and debugging abilities.

- [AI Model Efficiency Toolkit \(AIMET\)](#)

Open-source library to optimize (compressing and quantizing) trained neural network models. This is a complex SDK designed to generate optimized quantized models. It is intended only for advanced developers.

3 APIs

Qualcomm Linux provides [Qualcomm® Intelligent Multimedia SDK \(IM SDK\) Gstreamer plugin APIs](#) to interface with Qualcomm AI stack SDK APIs, to optimally run deep learning models on hardware modules such as NPU, GPU, CPU.



Three Gstreamer ML plugin APIs are provided to support two AI stack SDKs, and external TFLite, which provides flexibility for developers to choose the right combination for their AI needs.

Qualcomm IM SDK plugin for Qualcomm Neural Processing Engine (qtimlsnpe)	Uses Qualcomm Neural Processing Engine APIs to load and execute models. Choose this plugin for quick prototyping and high-level API support.
Qualcomm IM SDK plugin for Qualcomm AI Engine Direct plugin (qtimlqnn)	Uses Qualcomm AI Engine Direct APIs, which provide low-level, unified API and improved performance to optimize and execute network models on the desired hardware accelerator. Choose this plugin for advanced graph execution options and optimizations.
Qualcomm IM SDK plugin for TensorFlow Lite (qtimltflite)	Accelerates TFLite models directly using Qualcomm AI Engine Direct APIs to load and execute models. Choose this plugin to directly run TFLite models, without the need of conversion.

SDK APIs - These SDKs provide AI APIs for application developers.

Qualcomm Neural Processing Engine SDK	C, C++, and Java APIs to support heterogeneous computing, system-level configurations, and direct AI workloads to all accelerator cores. Provides developers with flexibility, including inter-core collaboration support and other advanced features.
Qualcomm AI Engine Direct	Low-level, highly customizable APIs that speed up AI models on all AI accelerator cores with individual libraries. This SDK can be used to target a specific accelerator core or delegate workloads from popular runtimes including Qualcomm Neural Processing Engine SDK, TensorFlow Lite, and ONNX runtime.

4 Sample Apps

Qualcomm Linux distribution provides sample AI/ML applications that demonstrate AI capabilities of the Qualcomm Linux platform.

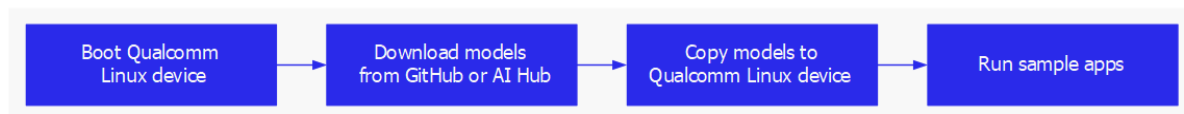


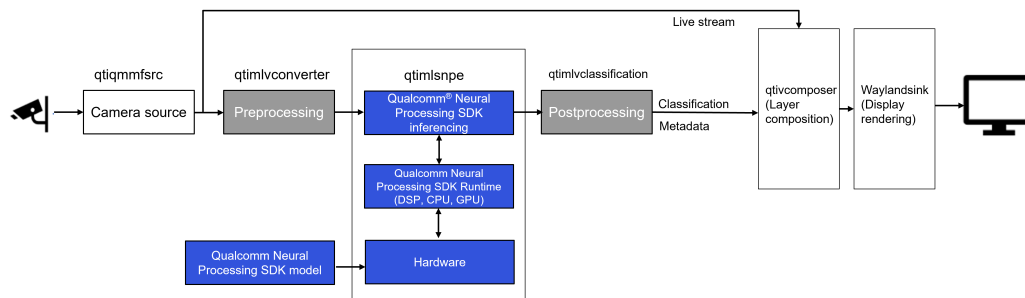
Figure1 Workflow for using the sample applications

The following AI sample applications are part of Qualcomm IM SDK.

Tip: For the complete list of supported sample applications, see [Sample applications](#) in the Qualcomm IM SDK documentation.

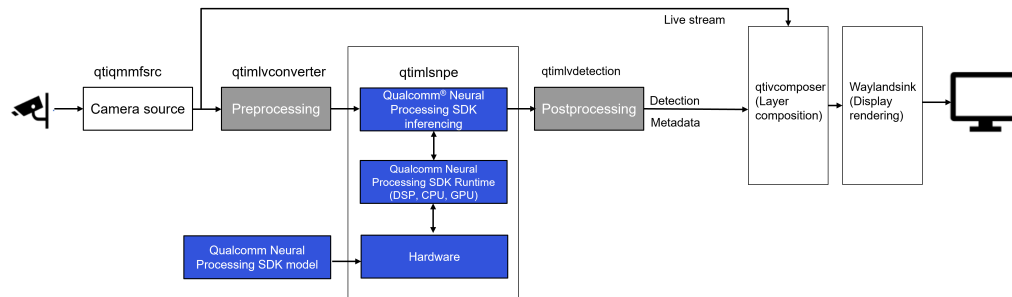
4.1 AI processing for object classification

The [gst-ai-classification](#) application demonstrates the capability of the hardware perform object classification on the camera live stream.



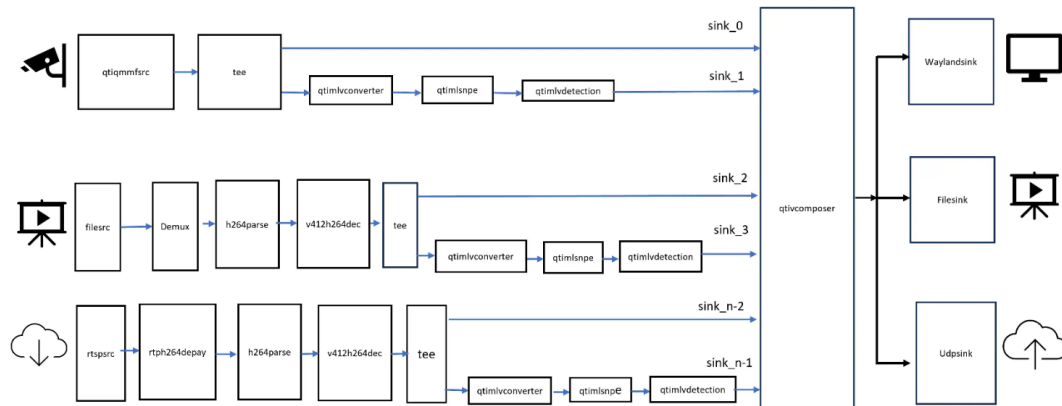
4.2 AI processing for object detection

The [gst-ai-object-detection](#) application demonstrates the capability of the hardware to perform object detection on camera live stream.



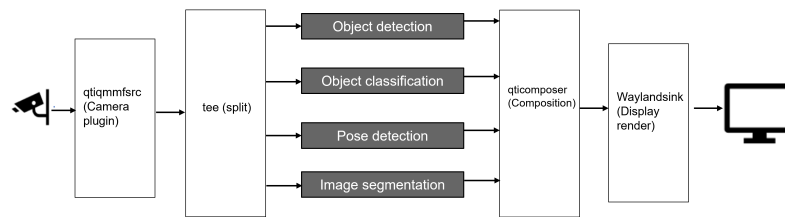
4.3 AI processing for multi-input/output object detection

The [gst-ai-multi-input-output-object-detection](#) application demonstrates the capability of the hardware to perform object detection on multiple video streams from camera, file, or Real-time Streaming Protocol (RTSP) source.



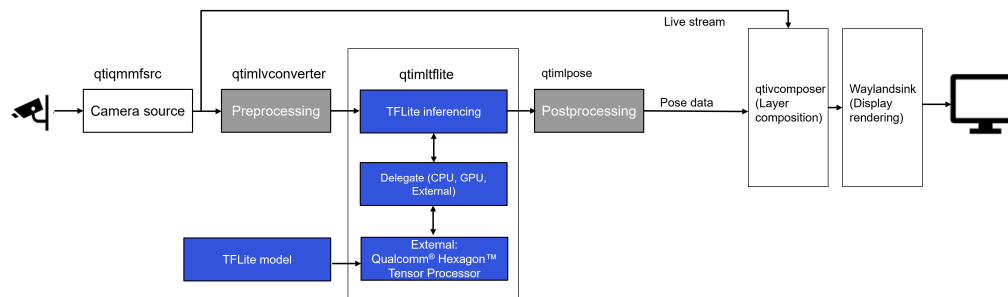
4.4 Parallel AI fusion

The [gst-ai-parallel-inference](#) application demonstrates the capability of the hardware to perform four parallel AI inferencing functions on the camera live stream.



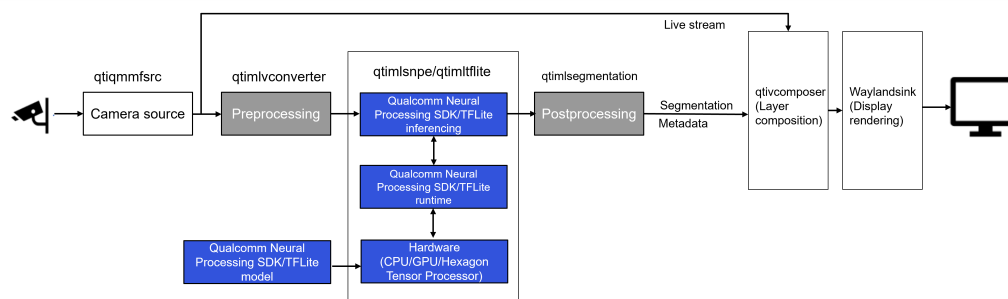
4.5 AI processing for pose detection

The [gst-ai-pose-detection](#) application demonstrates the capability of the hardware to perform pose detection on the camera live stream.



4.6 AI processing for image segmentation

The [gst-ai-segmentation](#) application demonstrates the capability of the hardware to perform image segmentation on the camera live stream.



5 Developer workflow

See the [complete AI Developer Workflow documentation](#)

6 References

Title	Number
AI Hub	—
Qualcomm Neural Processing Engine	80-63442-2
Qualcomm AI Engine Direct	80-63442-50
AI Engine Direct: TFLite Delegate	80-63442-50
Qualcomm Intelligent Multimedia SDK	80-70017-50

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